

SQL Set Operations: UNION, INTERSECT, and EXCEPT

Introduction

SQL Set Operations are used to combine the results of two or more SELECT statements. These operations allow users to compare, merge, and analyze data from different tables or queries. The three main set operations are:

1. UNION
 2. INTERSECT
 3. EXCEPT (called MINUS in Oracle)
-

1. UNION

Definition

The UNION operator combines the results of two or more SELECT statements into a single result set and removes duplicate rows.

When to Use

- When data is stored in different tables but has a similar structure.
 - When you want a combined list without duplicate values.
-

Syntax

```
SELECT column1, column2
FROM table1

UNION

SELECT column1, column2
FROM table2;
```

Rules & Conditions

1. Both SELECT statements must return the same number of columns.
2. Corresponding columns must have compatible data types.
3. Column names in the final result are taken from the first SELECT statement.
4. Duplicate rows are automatically removed.

Example (HR Schema)

Retrieve all employee IDs and department managers as one combined list.

```
SELECT EMPLOYEE_ID
FROM EMPLOYEES

UNION

SELECT MANAGER_ID
FROM DEPARTMENTS;
```

2. UNION ALL

Definition

UNION ALL combines results from multiple queries without removing duplicate rows.

Difference Between UNION and UNION ALL

Feature	UNION	UNION ALL
Removes duplicates	Yes	No
Performance	Slower	Faster
Returns all rows	No	Yes

Example

```
SELECT DEPARTMENT_ID
FROM EMPLOYEES

UNION ALL

SELECT DEPARTMENT_ID
FROM DEPARTMENTS;
```

If Department 90 appears multiple times, UNION removes duplicates while UNION ALL keeps all occurrences.

3. INTERSECT

Definition

INTERSECT returns only the rows that exist in both result sets.

When to Use

- To identify common records between two tables.
- To find matching values between datasets.

Syntax

```
SELECT column1  
FROM table1  
  
INTERSECT  
  
SELECT column1  
FROM table2;
```

Rules & Conditions

1. Both queries must return the same number of columns.
2. Data types must be compatible.
3. Duplicate rows are removed automatically.
4. Only common rows are returned.

Example (HR Schema)

Find department IDs that appear in both EMPLOYEES and DEPARTMENTS.

```
SELECT DEPARTMENT_ID  
FROM EMPLOYEES  
  
INTERSECT  
  
SELECT DEPARTMENT_ID  
FROM DEPARTMENTS;
```

Result:

- Returns only department IDs that exist in both tables.
-

4. EXCEPT / MINUS

Definition

EXCEPT returns rows from the first query that do not exist in the second query.

Oracle Note

Oracle Database uses MINUS instead of EXCEPT.

When to Use

- To find missing records.
 - To identify values existing in one table but not another.
-

Syntax (Standard SQL)

```
SELECT column1
FROM table1

EXCEPT

SELECT column1
FROM table2;
```

Syntax (Oracle)

```
SELECT column1
FROM table1

MINUS

SELECT column1
FROM table2;
```

Rules & Conditions

1. Both queries must return the same number of columns.
2. Data types must match.
3. Duplicate rows are removed automatically.
4. Returns only rows found in the first query and not in the second.

Example (HR Schema)

Find department IDs that exist in EMPLOYEES but not in DEPARTMENTS.

```
SELECT DEPARTMENT_ID
FROM EMPLOYEES

MINUS

SELECT DEPARTMENT_ID
FROM DEPARTMENTS;
```

Result:

- Displays department IDs referenced by employees but not defined in the DEPARTMENTS table.

Summary of Set Operations

Operation	Purpose	Removes Duplicates
UNION	Combines results from multiple queries	Yes
UNION ALL	Combines results from multiple queries	No
INTERSECT	Returns common rows from both queries	Yes
EXCEPT / MINUS	Returns rows from first query not in second query	Yes

Conclusion

Set operations are powerful SQL features that allow users to compare and combine data from multiple queries. UNION merges datasets, INTERSECT finds common records, and MINUS (EXCEPT) identifies differences between datasets. Understanding these operations is essential for reporting, data analysis, and enterprise database management.